

Théodore MICHEL

PhD in Computational Mathematics, High Performance Computing and Data

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Working Experience

- 2026-present **Research Associate** - POF, University of Twente, *Enschede, Netherlands*.
Sep-
 - 4D-Flow, blood flow reconstruction using ultrasound, microbubbles, and deep learning.
- 2026 **Research Associate** - ERC Cure, CEMEF Mines Paris PSL, *Sophia-Antipolis, France*.
Jan-Mar
 - Turbulent flow simulation surrogates using Graph Neural Networks coupled with Deep Reinforcement Learning
- 2022–2025 **PhD Candidate** - TSE énergie de confiance, *Bourgoin-Jallieu, France*.
Dec-Dec
 - Joint industrially-partnered PhD on agrivoltaic power plant digital twinning at Mines Paris PSL.
 - Large-scale wind flow computational fluid dynamics simulations, surrogates using graph neural networks, and optimization for decision-making using deep reinforcement learning.
 - 1 European patent, 3 publications (2 in submission process), 3 international conferences.
- 2023 **Visiting Researcher**, Farhat Research Group, Stanford University, *Palo Alto, USA*.
Oct-Nov
 - Subgrid modeling of supersonic parachute cables during inflation, under supervision of Prof. Charbel Farhat at Stanford Aeronautics & Astronautics department.
- 2022 **Climate Risk Analyst Intern** - AXA Climate, *Paris, France*.
Apr-Sep
 - Parametric insurance underwriting, risk analysis and stochastic modeling of Tropical Cyclones.
- 2021 **Research Intern** - IRD Géoazur, *Sophia-Antipolis, France*.
Jun-Jul
 - Earthquake early warning, magnitude estimation, and signal denoising using auto-encoders.
- 2020–2021 **Data Scientist Intern** - Solais, *Sophia-Antipolis, France*.
Dec-May
 - Development of a solar nowcasting tool for solar power plant using deep learning CNNs and sky imagers.
- 2020 **Research Intern** - CEMEF Mines Paris PSL, *Sophia-Antipolis, France*.
Jul-Nov
 - Shape and flow control optimization using adjoint for turbulent fluid dynamics problems.
- 2019 **Kitchen Clerk** - Elio, PWC Crystal Parc, *Levallois-Perret, France*.
Feb

Education

- 2026–2028 **Postdoc**, University of Twente, POF, ERC FALCON, Enschede, Netherlands.
Sep-Feb
- 2026 **Postdoc**, Mines Paris - PSL Research University, ERC Cure, Sophia-Antipolis, France.
Jan-Mar
- 2022–2025 **PhD in Computational Mathematics, High Performance Computing and Data**, Mines Paris - PSL Research University, Sophia-Antipolis, France.
Dec-Dec
Thesis: Towards a Digital Twin of a Solar Power Plant. Joint PhD (CIFRE) with TSE énergie de confiance.
- 2018–2022 **M.Sc. & Executive Engineering in Applied Mathematics, Artificial Intelligence & Big Data**, Mines Paris - PSL Research University, Paris, France.
Sep-Oct
Thesis: Climate Risk Analysis and Modeling of Tropical Cyclones. Grade: A.
Activities: Treasurer of Arts Association, Captain of English Debating Team (2 yrs)
- 2016–2018 **Preparatory School (B.Sc.) in Maths & Physics**, CPGE Lycée Joffre, Montpellier, France.
Sep-Sep
Preparation for France's selective "Grandes Ecoles" entry exams.
- 2014–2016 **International Baccalaureate**, Lycée Jules Guesde, Montpellier, France.
Jan-Jul
Anglo-American Section. Grade: 19.50/20 with Honors, valedictorian.

2006–2018 **Diploma of Musical Studies**, *Conservatoire de Montpellier*, Montpellier, France.
Musical Diploma for saxophone and Second Prize in music theory. Ensemble and chamber music.

Research and Technical Skills

Python Scientific computing, machine learning (Pytorch, scikit-learn, GPU); pre/post-processing workflows (VTK, PyQt); image processing.

C++ Development of solvers in FEM libraries (Cimlib); HPC applications (CPU, GPU), including parallel processing with MPI and CUDA.

Miscellaneous Version control and continuous integration with Git/GitLab; remote development via SSH; proficient in Linux; music production via Ableton Live

Languages

French (Native), English (Fluent, C2, TOEFL 116/120), Spanish (Fluent, C1), Chinese (Beginner, HSK2), Ancient Greek

Publication Work

1. **T. Michel**, A. Ansaldi, J. Viquerat, P. Meliga, E. Hachem. *Combining Machine Learning and Computational Fluid Dynamics for Solar Panel Tilt Angle Optimization in Extreme Winds*. Physics of Fluids (2024).
2. K. Juhel, Q. Bletery, **T. Michel** et al. *Fast and Full Characterization of Large Earthquakes from Prompt Elastogravity Signals*. Nature - Communications Earth & Environment (2024).
3. [In submission process] **T. Michel**, P. Meliga, E. Hachem et al. *ONECYL: a Computational Fluid Dynamics Dataset for Training Graph Neural Networks*. (2026).
4. [Pre-print] **T. Michel**, P. Meliga, E. Hachem. *Combining Graph Neural Networks and Deep Reinforcement Learning for Flow Control of Solar Panel Arrays*. (2026).

Conferences

DTE & AICOMAS Feb 2025 **IACM Digital Twins in Engineering Conference**, Paris. Podium presentation: *Graph Neural Networks for Computational Fluid Dynamics Space and Time Prediction: an Application to Solar Panels*.

WCCM Jul 2024 **World Congress on Computational Mechanics**, Vancouver. Podium presentation: *Deep Reinforcement Learning-Based Flow Optimization for Improved Wind Resilience of Solar Panel Arrays*.

ML4Fluids Feb 2024 **ERCOFTAC Workshop on Machine Learning for Fluids**, Paris. Podium presentation: *Flow Optimization using Deep Reinforcement Learning for Safety Preserving of Solar Panels Facing Wind Risks*.

Teaching Experience

2026 **Mines Paris - PSL**, Paris, France. Project Co-Host.
○ Course: Data-Challenge for fast and efficient surrogates of brain aneurysm blood flow simulations.

2023–2026 **Mines Paris - PSL**, Sophia Antipolis, France. Teaching Assistant.
○ Subjects: Computational Fluid Dynamics, Deep Reinforcement Learning, Graph Neural Networks.
○ Research Project Supervisor: lead 3 different research projects of Mines Paris students of 2 months each, in CFD and AI.

Talks and Seminars

June 2026 **University of Twente**, Enschede, Netherlands. Seminar for PoF group: *Deep Reinforcement Learning for Optimization in Turbulent Fluid Mechanics Problems and Perspectives*.

May 2024 **Université Nice Côte d'Azur**, Nice, France. *Talk for Doctoral School Forum: Building a Digital Twin of a Solar Power Plant: why and how?*

Projects

- 2019–present **An&mone**, Paris, France.
- Co-founder and member of electronic music group An&mone, art and design director, multiple published singles, albums and live concerts, community management.
- 2023–2024 **Deep-Tech Innovation and Entrepreneurship** Mines Paris - PSL, Paris, France.
- Sep-Jul ○ Followed courses of the Specialized Master on deep-tech innovation at Mines Paris PSL on entrepreneurship of research-based technical innovations (business model, startup, pitch, business plan).
- 2019–2020 **EducEco Challenge** Mines Paris - PSL, Paris, France.
- Sep-Aug ○ Project leader (part-time): solar-powered race car with team of 12 Mines Paris students and 12 advanced technicians in electronics and mechanics.

Prizes

- 2024 **Prix Pierre Lafitte** (finalist), Sophia Antipolis, France.
- PhD 2nd year award for excellence and innovation in partnership research.

Hobbies

Music and production, Cycling, Outdoors (trail, climbing), Painting, Writing, Triathlon

References

- **Prof. E. Hachem** - Mines Paris - PSL
- **Prof. C. Farhat** - Stanford University
- **Dr. Q. Bletery** - IRD Géoazur
- **Eng. J. Laborde** - TSE